

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO5	Assessment:	ASSESSMENT TOOL2- PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)
Weightage:	30%	Total Marks:	25
CO5 Statement:	Construct I/O communication mechanisms by comparing memory-mapped and I/O-mapped techniques, interrupt handling (vectored, hardware, software), DMA controllers, and advanced bus interfaces including PCIe, USB, NVMe, and Thunderbolt.		
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Department:	CSE (AI/ML)	Date:	

ANNEXURE A
Pseudocode Writing Task (Algorithm Design)

1. Write pseudocode to design a DMA-based data transfer algorithm that moves data from an I/O device to memory while handling interrupts and minimizing CPU involvement.
2. Develop pseudocode for an interrupt handling system that prioritizes vectored interrupts from multiple devices and dispatches appropriate service routines.
3. Write pseudocode to select between memory-mapped I/O and I/O-mapped I/O dynamically based on latency and bandwidth requirements.
4. Design pseudocode for a bus arbitration algorithm that manages multiple devices communicating over PCIe, USB, and Thunderbolt interfaces.
5. Write pseudocode to implement a hybrid I/O communication mechanism that uses polling for low-speed devices and DMA with interrupts for high-speed devices.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
 I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Problem Understanding (20%)	Algorithm Design (30%)	Use of Control Structures (20%)	Pseudocode Structure (10%)	Completeness (20%)
Q1	2	3	2	1	2
Q2	1.5	2	1.5	1	2
Q3	1.5	2.5	1.5	1	1.5
Q4	2	2	2	0.5	1.5
Q5	1.5	3	2	0.5	2

RUBRICS FOR CALCULATION:

43

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20% 2	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30% 3	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20% 2	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10% 1	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20% 2	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient